Overview

Wiki *Enhanced dissipation for passive scalars*

8 November 2026

Discuss on Zulip: #overview, topic "Overview".

[A short introduction to the topic, written as continuous prose: the phenomenon, the equation, why it matters, and what the reader will find in this wiki.]

$$\partial_t f + u \cdot \nabla f = \nu \Delta f, \qquad \nabla \cdot u = 0.$$

Video: `heat-vs-shear.mp4`, shown in the online edition.
Left: the heat equation. Right: the same initial datum carried by the shear flow $u = (\sin y, 0)$, with the same diffusivity. Bottom: the size of the two solutions in time, on a logarithmic scale. [numerics]

Video: `alternating-shears.mp4`, shown in the online edition.
Left: the shear flow $u = (\sin y, 0)$. Right: shears that alternate, horizontal and vertical, each acting for 3 units of time, with random phases. Same initial datum $f_0 = \cos x$, same diffusivity $\nu = 10^{-4}$; both velocity fields have size at most 1. Bottom: the size of the solutions in time, on a logarithmic scale. [numerics]

1 Map of the wiki

Diagram: `wiki-map.svg`, shown in the online edition.
The parts of the wiki; click a name to open its page.